

DIABETES READMISSION CLINICAL AI

EXECUTIVE REPORT

Development, Validation, Governance & Clinical Translation of a 30-Day Hospital Readmission Decision-Support Prototype

Application: ReadmitAI Clinical Intelligence

Domain: Healthcare / Clinical AI

Model: XGBoost | DIABETES_READMISSION_XGB_D13_V1 | v1.0.0

Validation: Internal Locked-Test Completed

Governance: Conditional Pass

Production Clinical Use: Not Authorized

1. EXECUTIVE SUMMARY

The Diabetes Readmission Clinical AI project demonstrates the end-to-end translation of a healthcare prediction problem into a validated, explainable, and governed clinical decision-support prototype.

The project addressed one question:

Can routinely available patient, admission-context, and prior-utilization information help identify diabetes-related hospital encounters warranting additional review for 30-day readmission risk?

The implementation covered the full clinical AI pathway:

Clinical Problem → Data → Machine Learning → Validation → Explainability & Robustness → Governance → Clinical Prototype → Monitoring

The analytical cohort contained 69,944 encounters involving 49,307 unique patients, including 7,907 30-day readmissions (11.30%).

Following development and validation, the selected XGBoost model was frozen at an operating threshold of 0.12 and evaluated once against 15,038 locked-test encounters.

Executive Evidence Snapshot

Measure	Result
Locked-Test Encounters	15,038
Readmissions / Prevalence	1,707 / 11.35%
ROC-AUC	0.623
PR-AUC	0.182
Sensitivity / Specificity	48.74% / 70.54%
PPV / NPV	17.48% / 91.49%
Alert Rate	31.65%
Number Needed to Evaluate	5.72

Performance remained broadly stable between validation and locked testing. However, assurance testing identified limited discrimination, material dependence on prior healthcare utilization, subgroup/admission-context heterogeneity, unresolved external transportability, and unestablished clinical effectiveness.

Governance Decision

CONDITIONAL PASS — PROGRESS WITH DOCUMENTED RESIDUAL RISKS

The model may progress through research-stage clinical translation and monitoring design. External validation and clinical effectiveness remain unestablished, and production clinical use is not authorized.

2. SOLUTION & CLINICAL TRANSLATION

The project used the UCI diabetes hospital readmission dataset containing 101,766 encounters and 50 source variables. Following data-quality assessment, leakage review, and governed cohort construction, 69,944 encounters formed the analytical cohort.

The final model uses 10 governed source features spanning patient characteristics, admission context, and prior healthcare utilization. Frozen preprocessing transforms these into 49 model-ready features.

Model Architecture

Clinical Inputs → Input Validation & Feature Engineering → Frozen Preprocessing → XGBoost Model → Readmission Probability → 0.12 Threshold → Prioritization Advisory → Explainability & Safety Context → Human Review

The system deliberately separates:

MODEL ESTIMATE — predicted readmission probability.

MODEL ADVISORY — prioritization signal.

CLINICAL DECISION — remains with the healthcare professional.

It is not designed to autonomously determine diagnosis, treatment, discharge or access to care.

The validated model was subsequently translated into ReadmitAI Clinical Intelligence, a research prototype integrating:

Risk Assessment | Explainability | Model Evidence | Governance & Safety | Monitoring

3. VALIDATION, SAFETY & GOVERNANCE

The frozen model underwent one-time confirmatory locked-test evaluation without post hoc retraining, feature redesign or threshold optimization using locked-test outcomes.

No material aggregate performance collapse was observed relative to preceding validation. More importantly, deeper assurance identified clinically relevant limitations.

Key Assurance Finding — Utilization Dependence

Prior healthcare utilization provided important predictive signal but also became a material model dependency.

Among the evaluated locked-test profile with zero recorded prior outpatient, emergency, and inpatient utilization, sensitivity at the frozen threshold was 0%.

This was translated directly into a runtime safety control:

Absence of a priority flag in patients with no recorded prior utilization must not be interpreted as absence of readmission risk.

SHAP analysis independently reinforced this finding: utilization-related features accounted for approximately 68.52% of validation attribution and 77.27% of locked-test attribution. SHAP results are explicitly governed as model contribution—not clinical causality.

Residual Risk Position

Risk	Governance Position
Utilization-Dependent Behavior	MATERIAL
Explainability Utilization Dominance	MATERIAL
Limited Model Discrimination	MATERIAL
Subgroup / Admission-Context Heterogeneity	DOCUMENTED
External Transportability	UNRESOLVED
Clinical Effectiveness	UNRESOLVED

The model therefore received a Conditional Pass, not unrestricted approval.

This distinction is fundamental: successful internal validation does not equal clinical effectiveness or production authorization.

4. DEPLOYMENT, MONITORING & EVIDENCE POSITION

ReadmitAI was deployed as a research and validation prototype, not as a production clinical system.

The monitoring design contains 17 controls covering data and schema quality, missingness, input/prediction drift, alert rates, discrimination, calibration, operating-point performance, subgroup performance, utilization dependence, admission context, explainability stability, clinical overrides, workflow adoption and model incidents.

Escalation is structured as:

CRITICAL → ALERT → WATCH → NOT ASSESSABLE

Current Evidence Position

Evidence Domain	Status
Internal Locked-Test Validation	COMPLETED
Explainability & Robustness	COMPLETED WITH RESIDUAL RISKS
Subgroup/Fairness Assessment	HETEROGENEITY DOCUMENTED
External Validation	NOT ESTABLISHED
Prospective Clinical Validation	NOT ESTABLISHED
Clinical Effectiveness	NOT ESTABLISHED

The next evidence pathway is:

External Validation → Transportability & Calibration → External Subgroup/Robustness Assessment → Prospective Clinical Evaluation → Clinical Effectiveness → Production Governance Decision

5. EXECUTIVE CONCLUSION

The Diabetes Readmission Clinical AI project demonstrates substantially more than predictive model development.

It integrates:

Healthcare Machine Learning | Clinical Model Validation | Explainable AI | Robustness & Fairness | Responsible AI Governance | Model Risk Management | Human Oversight | Clinical AI Translation | Monitoring

The project also demonstrates a critical principle of responsible healthcare AI:

Model limitations should become governance controls—not disappear behind aggregate performance metrics.

The final evidence position is therefore intentionally explicit:

INTERNAL VALIDATION — COMPLETED
GOVERNANCE — CONDITIONAL PASS
EXTERNAL VALIDATION — NOT ESTABLISHED
CLINICAL EFFECTIVENESS — NOT ESTABLISHED
PRODUCTION CLINICAL USE — NOT AUTHORIZED

The project demonstrates the ability to move from clinical problem definition to working AI implementation while maintaining evidence boundaries, clinical safety, and governance discipline.

Capability Demonstrated:
Clinical AI | Healthcare Data Science | Clinical Model Validation | Responsible AI Governance | Clinical AI Translation

DOCUMENT CONTROL

Field	Status
Document	Executive Report — Final
Model	v1.0.0
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Governance	Conditional Pass
External Validation	Not Established
Clinical Effectiveness	Not Established
Production Clinical Use	Not Authorized